PES UNIVERSITY

Department of Computer Science and Engineering

Software Engineering

LAB 1

Problem Statement #13 – Healthcare & Telemedicine

Patient Health Record Consent Management System

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1. Problem Context & Overview

A patient-centric electronic health data gateway where patients explicitly manage granular, time-bound consent permissions for clinics, diagnostic labs, and consulting doctors to access their medical history.

Target Stakeholders / Actors: Patient, Clinic Administrator

7. Use-Case Flow Specification

7.1 Preconditions

1. The patient has access to the system.
2. The patient is authenticated.
3. The healthcare provider/doctor is verified.
4. The patient's health records are available in the system.
5. The patient has selected the records that can be shared.

7.2 Postconditions

6. The selected consent is stored in the system.
7. The selected healthcare provider can access the permitted records.
8. The consent contains an expiry time.
9. The consent grant event is recorded in the audit trail.
10. Access automatically stops when the consent expires.

7.3 Main Success Scenario

11. Patient logs into the Patient Health Record Consent Management System.
12. The system authenticates the patient.
13. Patient selects Grant Consent.
14. The system displays the list of available health/diagnostic records.
15. Patient selects the records that can be shared.
16. Patient selects the verified doctor or healthcare provider.
17. Patient enters or selects the consent duration, for example 24 hours.
18. The system displays the selected records, healthcare provider and expiry time for confirmation.
19. Patient confirms the consent.
20. The system creates the consent permission.
21. The system records the consent grant event in the audit trail.
22. The system displays a confirmation message to the patient.
23. The healthcare provider can now access only the selected records while the consent is active.
24. When the expiry time is reached, the system automatically removes the access permission.
25. The use case ends successfully.

7.4 Alternate Flow A1 – Doctor Is Not Verified

26. Patient selects a doctor to give access.
27. The system checks the doctor's verification status.
28. The doctor is found to be unverified.
29. The system does not allow the patient to grant access to that doctor.
30. The system displays a message: “Selected doctor is not verified.”
31. Patient can select another verified doctor.
32. The use case continues from the provider-selection step.

7.5 Alternate Flow A2 – Patient Cancels Consent

33. Patient selects the records and healthcare provider.
34. The system displays the consent details.
35. Patient chooses Cancel instead of confirming.
36. The system does not create the consent.
37. No access is given to the healthcare provider.
38. The use case ends.

7.6 Alternate Flow A3 – Patient Revokes Consent Before Expiry

39. Patient opens the active consent list.
40. Patient selects an active consent.
41. Patient chooses Revoke Consent.
42. The system asks for confirmation.
43. Patient confirms revocation.

44. The system immediately disables the provider's access.
45. The system records the revocation event in the audit trail.
46. The system displays a confirmation message.
47. The use case ends.